

Statistical and Machine Learning SS26

Exercise Sheet 7

Before we start

Installation of Python

- Windows users – you can install Python on Windows following, for example, this YouTube video.
- Mac users – you can install Python on Mac, following, for example, this YouTube video.

Using IDE (Integrated Development Environment)

You can run Python directly from Run (Windows) or Termina (Mac). But running it from an IDE offers a smoother software-user interaction. There are many IDE you can run Python with. A few example include:

- IDLE (comes with Python installation)
- PyCharm
- RStudio
- Spyder
- Sublime
- Visual Studio Code
- ... and many, many others.

You are free to use IDE of your choice. For the ease of presentation, all the examples in this class will be showed using Jupyter Notebook.

Instructions on how to install Jupyter Notebook can be found below:

Installing Jupyter Notebook on Mac and Windows

Jupyter Notebook is a browser-based tool for writing and running Python code.

1. Check that Python and pip are installed

Mac

Open **Terminal** and run:

```
python3 --version
python3 -m pip --version
```

Windows

Open **Command Prompt** or **PowerShell** and run:

```
python --version
python -m pip --version
```

If Python is not installed, download it from:

<https://www.python.org/downloads/>

On Windows, make sure to select **Add Python to PATH** during installation.

2. Create a project folder

Mac

```
mkdir ~/python-practice
cd ~/python-practice
```

Windows Command Prompt

```
mkdir %USERPROFILE%\python-practice
cd %USERPROFILE%\python-practice
```

Windows PowerShell

```
mkdir $HOME\python-practice
cd $HOME\python-practice
```

3. Create a virtual environment

A virtual environment keeps Python packages organized for one project.

Mac

```
python3 -m venv .venv
source .venv/bin/activate
```

Windows Command Prompt

```
python -m venv .venv
.venv\Scripts\activate
```

Windows PowerShell

```
python -m venv .venv
.venv\Scripts\Activate.ps1
```

If PowerShell blocks activation, run:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

Then activate the environment again:

```
.venv\Scripts\Activate.ps1
```

4. Upgrade pip

Mac

```
python3 -m pip install --upgrade pip
```

Windows

```
python -m pip install --upgrade pip
```

5. Install Jupyter Notebook

Mac

```
python3 -m pip install notebook
```

Windows

```
python -m pip install notebook
```

6. Start Jupyter Notebook

Mac

```
python3 -m notebook
```

Windows

```
python -m notebook
```

A browser window should open automatically. From there, create a new Python notebook.

7. Starting Jupyter later

Each time you want to use Jupyter again, go to your project folder and activate the virtual environment.

Mac

```
cd ~/python-practice  
source .venv/bin/activate  
python3 -m notebook
```

Windows Command Prompt

```
cd %USERPROFILE%\python-practice  
.venv\Scripts\activate  
python -m notebook
```

Windows PowerShell

```
cd $HOME\python-practice  
.venv\Scripts\Activate.ps1  
python -m notebook
```

8. Closing Jupyter Notebook

To fully close Jupyter Notebook, do not only close the browser tab. Closing the browser tab closes the page, but the Jupyter server may still be running.

Go back to the **Terminal**, **Command Prompt**, or **PowerShell** window where Jupyter is running.

Press:

```
Control + C
```

If you see this message:

```
Shutdown this Jupyter server (y/[n])?
```

type:

```
y
```

Then press **Enter**.

Jupyter Notebook is now closed.